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Some experiments on adiabatic shear failure

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Abstract. Dynamic testing of statically pre-deformed specimens of Ti6Al4V shows that the total strain to failure increases with the level of pre-strain. Furthermore, interrupted and various other dynamic tests on Ti6AL4V show that the temperature rise prior to localization has only a minor influence on adiabatic shear band (ASB) formation. For all the tests, the calculated dynamic deformation energy until ASB formation is found to be relatively constant. This indicates that adiabatic shear banding is dependent almost solely on dynamic deformation processes, with quasi-static and thermal effects prior to localization being very marginal. We suggest the concept of a constant dynamic mechanical energy (toughness) as a quantitative criterion for ASB formation. Physically, this concept is related to the dynamic stored energy of cold work

Adiabatic shear failure is a well know failure mechanism that is uniquely experienced in dynamic loading situations [1, 2]. The homogeneous deformation tends to localize into a narrow band followed by catastrophic failure. Because of the typical short time scales involved, the phenomenon is considered to be adiabatic with large temperature increases, especially inside the band. Many materials fail by adiabatic shear banding, in an uncontrolled manner. A vast body of literature has been dedicated to the material aspects of adiabatic shear failure [1, 2]. A basic criterion has been proposed by Zener and Hollomon [3], stating that ASB is governed by the competition between strain hardening and thermal softening of the material, so that the critical conditions for ASB formation are met when the material looses its strain hardening capacity. This criterion has been the basic ingredient for the vast majority of subsequently proposed criteria. These criteria were developed for specific constitutive models, and it can be noted that they all point to a critical strain value for the onset of the unstable failure process [2, 4]. By contrast with the ample literature dedicated to the material aspects of ASB's, there are only a few experimental attempts to investigate the mechanical aspects of the problem. Noticeable exceptions are the work of Marchand and Duffy [5] on dynamic torsion with simultaneous temperature measurements, and also work by Johnson and Cook [6] who verified the validity of their constitutive model for the prediction of the critical strain to failure. Other experimental investigations were aimed at provoking controlled shear band nucleation from a crack-tip, as in Zhou et al. [7, 8], and in this context, the concept of shear band toughness was invoked.

The purpose of the present work is to bring additional experimental evidence on adiabatic shear banding from a mechanical point of view, *emphasizing the effects of the prior mechanical history of the material (prestrain) and thermo-mechanical aspects of the localization process.*

The selected material is Ti6Al4V, annealed for 2 hours at 1200 F and air cooled, resulting in a mixed α-β microstructure. The material was supplied as 12 mm diameter rods. Quasi-static tests were performed on a servo-hydraulic testing machine (MTS 810), while high strain rate tests were performed on a 19 mm diameter split Hopkinson pressure bar [9]. All the tests were carried out on shear compression specimens (SCS) [10-14].

Figure 1 presents typical stress-strain curves obtained for quasi-static and dynamic loading. The typical quasi-static failure strain is $\varepsilon_{FS} \approx 0.24$, while for dynamic loading, $\varepsilon_{FD} \approx 0.17$. The SCS specimens failed by adiabatic shear at high strain rates. The dynamic final failure stage is quite abrupt. Thus, once a shear band has nucleated, it propagates and fractures very rapidly. This well-defined initiation stage makes Ti6Al4V an ideal model material for ASB studies.

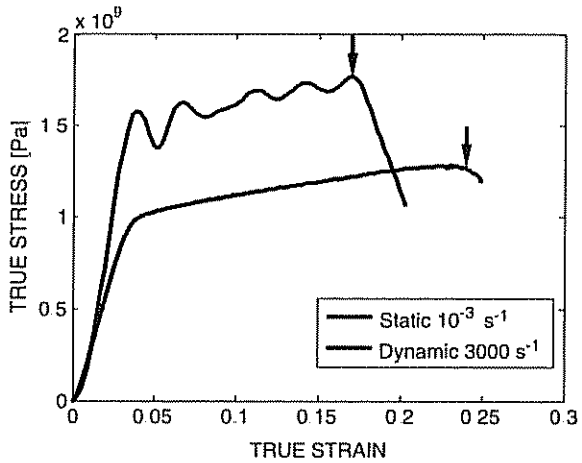


Figure 1. Typical quasi-static and dynamic stress-strain curves of Ti6Al4V alloy (arrows indicate failure strain)

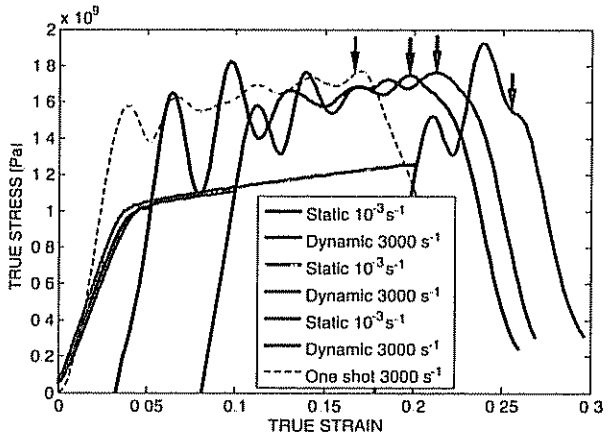


Figure 2. Quasi-static preload tests followed by dynamic tests (arrows indicate failure strain)

Two sets of experiments were performed: "static-dynamic" and "dynamic at different initial temperature". The "static-dynamic" tests started with static preloading to a predetermined level of strain, followed by dynamic loading. Figure 2 shows that the total (static and dynamic) strain to failure ε_{tot} increases noticeably with the level of pre-strain. This is contrary to the constant failure strain criterion requiring failure strain to remain constant. This strain may reach $\varepsilon_{tot} \approx 0.26$ and is larger than the characteristic failure strains obtained from pure quasi-static and dynamic tests ($\varepsilon_{FS} \approx 0.24$, $\varepsilon_{FD} \approx 0.17$ respectively). An additional quantitative insight is gained by calculating the dynamic mechanical energy involved until failure, i.e. the area of the dynamic stress-strain curve up to instability (indicated by the arrows in Figures 1 and 2). The result (Figure 4-a) shows that the dynamic mechanical energy is *practically constant*, irrespective of the initial pre-strain levels, up to very large pre-strain levels ($\varepsilon_{tot} \approx 0.15 = 0.63\,\varepsilon_{FS}$). Beyond this level, the drop in mechanical energy remains finite($\approx 50\%$).

The results presented here show that the widely accepted concept of a critical strain cannot strictly predict the onset of adiabatic shear failure, if the overall mechanical history of the material is to be taken into account. We propose an alternative criterion to identify the onset of ASB, using the value of the overall dynamic mechanical energy up to catastrophic failure, at which it reaches a critical value (toughness). For various large static pre-strain levels, the dynamic mechanical energy is remarkably

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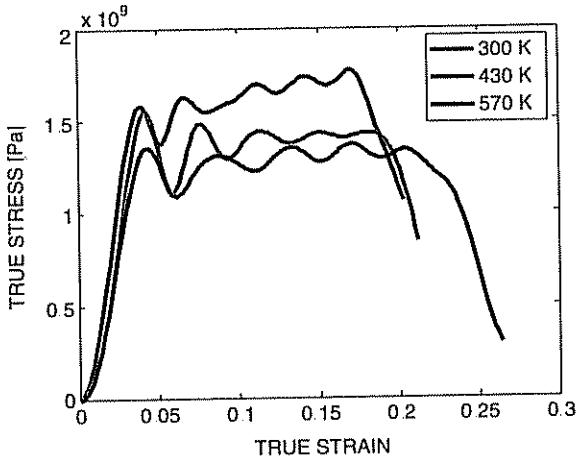


Figure 3. Ti6Al4V dynamic stress-strain curve at different initial temperature ($\dot{\varepsilon} = 3000/s$).

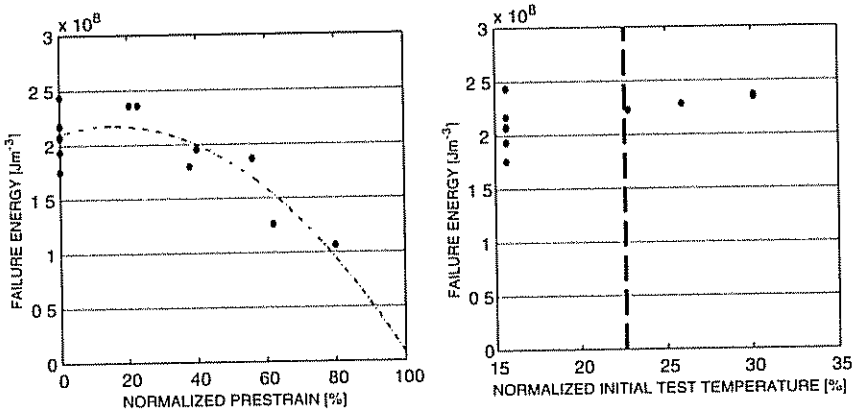


Figure 4. a) Dynamic mechanical energy vs. the normalized prestrain $\varepsilon_{pre}/\varepsilon_{FS} \times 100$. **b)** Dynamic mechanical energy vs. the normalized initial test temperature $T_{test}/T_{melt} \times 100$. The dashed line indicates the maximum calculated homogeneous temperature rise.

constant. One could therefore postulate that this critical mechanical energy is a material property (toughness) which can easily be measured and implemented in a numerical procedure.

One can assess the influence of thermal coupling effects *prior to localization* by calculating the maximum homogeneous temperature rise prior to localization. Using typical values of $\sigma = 1500$ MPa, $\varepsilon = 0.17$, $\rho = 4430\,kg/m^3$, $Cp = 526\,J/kg \times K$, and $T_{melt} = 1873$ K, the *maximum* temperature rise in the gage is found to be $T_{max} = 410$ K, corresponding to a homologous temperature $T_{hom} = 0.22$ [15, 16]. Such a temperature is not negligible and the material is expected to undergo some thermal softening.

A series of dynamic tests was carried out at various initial temperatures, up to 570 K. Figure 3 shows typical dynamic stress-strain curves and Figure 4-b shows the corresponding dynamic mechanical (failure) energy. It can be seen that, as the initial test temperature increased, the material's failure characteristics were not markedly affected. The stress decreases only slightly at the higher test temperature, while the failure energy is almost not affected by temperature in the investigated range.

One should note that the high temperature tests (Figures 3 and 4-b) reached a homologous temperature of $T_{hom} = 0.3$ (actual T=570 K) that is *far higher* than any temperature reached during the homogeneous stage of the dynamic deformation. Yet, the dynamic mechanical energy remained relatively unaffected.

All these observations suggest that the specimen's temperature *prior to adiabatic shear localization* has only a very minor effect on the onset of ASB formation, in agreement with results of Clos et al. [17] who report a minor temperature rise prior to localization in their steel specimens. From a practical point of view, thermal effects can be verified experimentally through real time monitoring of the specimen's temperature, a point under current investigation. These results also emphasize the importance of the stored energy of cold work, which is *precisely the fraction of energy that is not dissipated into heat*. The fact that ASB formation is apparently totally dependent on the dynamic stored energy of cold work suggests that relevant physical mechanisms are most likely related to the specific microstructure (e.g. dislocation patterns and hardening) that results from dynamic loading only, in agreement with the observations of Kennedy and Murr [18].

Therefore, the main conclusions of the present study are:

- The concept of a critical strain for ASB formation does not accurately reflect the overall mechanical history of a material.
- The homogeneous specimen temperature (either adiabatic rise or from test conditions) prior to strain localization has a very minor influence on the subsequent ASB formation.
- The dynamic mechanical energy has a constant value (toughness) at the onset of ASB formation, for a large range of quasi-static pre-strain levels and initial test temperatures.
- The dynamic stored energy of cold work is experimentally identified as the governing factor for ASB formation.
- This constant energy reflects a state beyond which the material's microstructure can no longer be modified, so that all the mechanical energy goes to heat. As such, this point can be identified experimentally.

Acknowledgement

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